

Assessing the effectiveness of the NIST DART-MS Forensics Database and Data Interpretation Tool for designer drug screening with alternative instrumentation

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ABSTRACT

The increasing prevalence of designer drugs has contributed towards the adoption of rapid screening techniques, such as direct analysis in real time-mass spectrometry (DART-MS). Recently, the National Institute of Standards and Technology (NIST) developed a DART-MS Forensics Database and Data Interpretation Tool (DIT) to assist practitioners with the interpretation of seized drug mixtures. However, this resource was only designed for DART ionization coupled with AccuTOF high-resolution mass spectrometers. This study assesses the effectiveness of the NIST DART-MS Forensics Database and DIT for mixture interpretation using other soft ionization sources, such as electrospray ionization (ESI), and coupling DART to an Agilent 6530 quadrupole time-of-flight (Q-TOF) mass spectrometer. As the seized drug community continues to shift towards the use of in-source collision-induced dissociation (IS-CID) with single-stage mass spectrometers coupled with soft ionization sources, it is imperative to understand the similarity of the generated IS-CID fragment ion spectra and the effectiveness of the DIT for seized drug screening using instrumentation other than the DART-AccuTOF.

Pearson product-moment correlations (PPMCs) revealed a high degree of spectral similarity between the ESI-IS-CID and DART-IS-CID fragment ion spectra collected across a range of activation conditions. The 95% confidence interval for the average highest PPMC coefficient across the 12 designer drugs in this study was 0.9295 ± 0.0031 . More importantly, nearly identical mixture interpretation results were obtained when the ESI-IS-CID and DART-IS-CID fragment ion spectra were analyzed with the DIT. In all cases, the DIT was able to identify a potential library match with at least two match types between the low and high IS-CID spectra for all compounds present in each mixture. Demonstrating the potential applicability of the NIST DART-MS Forensics Database and DIT for the screening of seized drug mixtures using instrumentation other than the DART-AccuTOF provides support for the expanded applicability of this freely available resource.

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1. Introduction

The Controlled Substances Act of 1970 established a structured way for substances to be scheduled based on their potential for abuse, dependence, and accepted medical use [1]. The establishment of this act has led to the rise in the production of designer drugs [2]. Designer drugs are produced by incorporating slight structural modifications to a controlled substance to avoid current regulations [3–5]. The intention of designer drugs is to provide a

“legal high” through pharmacological effects similar to or greater than the original controlled substance [6,7]. These synthetic compounds are typically created in clandestine laboratories with limited regulation and quality control, often resulting in the formation of dangerous substances and hazardous byproducts or impurities [3]. Despite the known dangers, abuse of designer drugs continues due to their unscheduled nature and the increased difficulty for identification by routine drug screening methods.

Synthetic cannabinoids, synthetic cathinones, and fentanyl analogs are three of the most encountered designer drug classes. Synthetic cannabinoids provide marijuana-like pharmacological effects to the user [8]. Although structurally similar to Δ^9 -tetrahydrocannabinol (Δ^9 -THC), which is the main psychoactive

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chemical constituent of *Cannabis sativa* L., synthetic cannabinoids have the potential to bind to the CB1 and CB2 receptors with an even greater affinity than their scheduled counterpart [9–11]. In comparison, synthetic cathinones provide stimulant-like effects due to their core cathinone-like structure [12–14]. Cathinone is the psychoactive constituent of the *Catha edulis* plant, commonly referred to as khat, that is chewed across Africa and the Arabian Peninsula for stimulant-like effects [15,16]. Synthetic cathinones are commonly labeled as “bath salts,” “plant food,” or “research chemicals” to further avoid regulations [17]. Both synthetic cannabinoids and synthetic cathinones are scheduled on an analog-by-analog basis due to the inherent hazards to society as determined by the Drug Enforcement Administration (DEA). Consequently, the scheduling of synthetic cannabinoids and synthetic cathinones cannot keep up with the constant influx of novel designer drugs [18]. In contrast, all fentanyl-related substances were temporarily placed into Schedule I of the Controlled Substances Act in February 2018 [19], which has since been extended due to the potential of fentanyl analogs to produce respiratory and central nervous system depression [20].

With the continued influx of designer drugs, the seized drug community has moved towards the incorporation of soft ionization sources coupled with single-stage mass spectrometers. In particular, soft ionization techniques such as ESI and DART, which provide enhanced molecular weight information through the formation of predominantly protonated/deprotonated molecules [2,21], have become increasingly popular. However, soft ionization sources typically only provide molecular weight information unless combined with an activation technique, which is limited to in-source collision-induced dissociation (IS-CID) on single-stage mass spectrometers. IS-CID involves collisional activation of ions during the transition from the atmospheric pressure ionization source to the high vacuum of the mass analyzer region. The typical voltages used for ion transmission can be increased such that ions undergo energetic collisions with residual background gases, leading to a controllable degree of fragmentation [22–24]. Examples of instrumentation where IS-CID can be used to gain structural information include liquid chromatography-electrospray ionization-mass spectrometry (LC-ESI-MS) and DART-mass spectrometry (DART-MS). LC-ESI-MS with IS-CID activation offers similar fragmentation to tandem mass spectrometry (MS/MS) instrumentation at a more affordable cost [25]. In comparison, DART-MS offers higher throughput screening analyses due to the limited sample preparation required prior to analysis [26,27].

The combination of soft ionization techniques and IS-CID activation has been demonstrated previously for the identification of synthetic cannabinoids [25,28], synthetic cathinones [21,29,30], and fentanyl analogs [29,31], although only with manual interpretation. Recently, NIST developed a publicly available DART-MS Forensics Database [32] and Data Interpretation Tool (DIT) [33,34] to assist the seized drug community with the identification of unknown substances, such as designer drugs. The DART-MS Forensics Database currently contains IS-CID spectra at low, medium, and high activation conditions for more than 1100 seized drugs or seized drug-related compounds. The DIT allows users to upload spectra to compare against the DART-MS Forensics Database and contains a novel algorithm, known as the inverted library search algorithm (ILSA), to assist analysts with mixture interpretation based on the uploaded IS-CID fragment ion spectra [33,34].

However, the data comprised within the DART-MS Forensics Database was collected with DART ionization coupled to JEOL AccuTOF high-resolution mass spectrometers only, and this freely available resource has not been applied to the identification of seized drugs using IS-CID activation with other soft ionization sources or with DART ionization coupled to other high-resolution

mass spectrometers. This study provides the first assessment of the effectiveness of the NIST DART-MS Forensics Database and DIT for mixture interpretation when using other soft ionization sources or a different high-resolution mass spectrometer through the collection of IS-CID data for representative designer drugs using both ESI and DART coupled to an Agilent 6530 quadrupole time-of-flight (Q-TOF) mass spectrometer. The similarity of IS-CID fragment ion spectra generated with ESI and DART ionization was assessed using spectral comparisons and Pearson product-moment correlations (PPMCs). A PPMC coefficient provides a statistical assessment of the strength and orientation of the correlation between two input vectors (i.e., fragment ion spectra), with output values closer to 1 indicating higher spectral similarity [35]. The resulting ESI-IS-CID and DART-IS-CID fragment ion spectra were uploaded to the DIT and searched against the NIST DART-MS Forensics Database to determine the effectiveness of this approach for the identification of the representative designer drugs. Assessing the effectiveness of the NIST DART-MS Forensics Database and DIT for the identification of seized drug mixtures using instrumentation other than the DART-AccuTOF will inform the seized drug community about the applicability of this freely available resource to routine seized drug screening for non-DART-AccuTOF users.

2. Methods

2.1. Chemicals and materials

The representative designer drugs chosen for this study included 12 analytical reference standards obtained from Cayman Chemical (Ann Arbor, MI, USA) or Cerilliant (Round Rock, TX, USA). The synthetic cannabinoids consisted of AB-FUBINACA, AB-PINACA, MAB-CHMINACA, and AM-2201. The synthetic cathinones consisted of 3-methoxymethcathinone (3-MeOMC), 4-chloroethcathinone (4-CEC), pentylone, and α -pyrrolidinovalerophenone (α -PVP). The fentanyl analogs consisted of fentanyl, alfentanil, α -methyl acetyl fentanyl, and crotonyl fentanyl.

2.2. Sample preparation

All samples were prepared in a 49.9% HPLC grade methanol (J.T. Baker, Phillipsburg, NJ, USA), 49.9% deionized water, and 0.2% glacial acetic acid (Mallinckrodt, St. Louis, MO, USA) solvent. An in-house Millipore Direct-Q® 3 UV system (Burlington, MA, USA) was used to generate the necessary 18.2 M Ω deionized water. The synthetic cannabinoids and synthetic cathinones were prepared and analyzed as 50 ppm solutions, whereas the fentanyl analogs were prepared and analyzed as 10 ppm solutions to mitigate the increased risks related to working with fentanyl analogs. The working solutions were prepared in 2 mL microcentrifuge tubes (Fisher Scientific, Waltham, MA, USA) and stored in a freezer (–20 °C) to prevent analyte degradation.

Along with the analysis of the pure compounds, five simulant mixtures were created containing between two and four designer drugs. The mixtures contained: 1) α -methyl acetyl fentanyl and fentanyl; 2) 3-MeOMC and AB-FUBINACA; 3) AM-2201, 4-CEC, and alfentanil; 4) MAB-CHMINACA, α -PVP, and crotonyl fentanyl; and 5) AB-PINACA, fentanyl, crotonyl fentanyl, and pentylone. The synthetic cathinone, synthetic cannabinoid, and fentanyl analog mixtures were prepared at the same 50 ppm and 10 ppm concentrations, respectively, in 2 mL microcentrifuge tubes and stored in a freezer (–20 °C) to prevent analyte degradation.

2.3. Instrumentation

The pure compounds and mixtures were analyzed with an

Agilent Technologies 6530 Q-TOF mass spectrometer. For the ESI analysis, a dual Agilent Jet Stream (AJS) electrospray ionization source was operated in positive ionization mode with a capillary voltage of 3500 V, a nozzle voltage of 1500 V, a 300 °C drying gas at a flow rate of 8 L/min, a nebulizer pressure of 40 psi, and a 350 °C sheath gas at an 8 L/min flow rate as described previously [25]. The nitrogen used for the ESI source was generated in-house with a Genius 3040 nitrogen generator (Peak Scientific, Billerica, MA, USA). A 15 μ L/min flow rate was used for direct injection sample introduction for the ESI analysis. The DART JumpShot source (Ion-Sense, Saugus, MA, USA) was operated in positive ionization mode, using high-purity helium as the source gas, with a source gas temperature of 300 °C, and an exit grid voltage of 350 V. The DART sample preparation included the deposition of 3 μ L of sample onto the closed end of a glass capillary (Corning Inc., Corning, NY, USA), allowing the deposited sample to dry, and then introducing the sample into the DART source for approximately 10 s. All samples for both ESI and DART ionization were collected with a scan range of m/z 40–450. The voltages used for IS-CID activation were the fragmentor and skimmer voltage. The fragmentor and skimmer voltages used for the ESI-IS-CID activation ranged from 150 V to 350 V in increments of 10 V with a constant skimmer voltage of 65 V. The DART-IS-CID fragmentor voltages were first optimized for the targeted low, medium, and high activation conditions. Fragmentor voltages of 200 V (low), 350 V (medium), and 380 V (high) and a skimmer voltage of 65 V were used for the DART-IS-CID activation. Only three IS-CID activation conditions were studied with the DART-IS-CID to parallel the input of low, medium, and high IS-CID activation data into DIT.

2.4. Data analysis

Mass spectral data was extracted through MassHunter Qualitative Analysis version B.08.00 (Agilent Technologies, Santa Clara, CA, USA) as both profile and centroid data, then exported as .CSV files into Microsoft Excel version 16 (Microsoft Corporation, Redmond, WA, USA). The profile data was background subtracted using the average combined blank spectrum collected from a blank capillary, a solvent blank, and a microcentrifuge tube blank for the low, medium, and high DART-IS-CID activation conditions. This profile data was then used to assess the statistical similarity of the ESI- and DART-generated IS-CID fragment ion spectra under low, medium, and high IS-CID activation conditions using PPMCs. The PPMC comparisons were computed from m/z 40 to the corresponding m/z value for the protonated precursor plus 4 Da (i.e., $[M+5]^+$) as described previously [25]. This comparison range was chosen to include the presence of any isotopes, but to avoid the presence of adducts that are expected to be present for ESI more so than for DART. The NIST DART-MS Forensics Database and DIT (version 2.0d) were used with the collapsed Firefly library for the analysis of the simulant mixtures with ESI- and DART-generated IS-CID fragment ion spectra.

3. Results and discussion

3.1. Spectral comparisons between IS-CID fragment ion spectra

Spectral comparisons between ESI- and DART-generated IS-CID fragment ion spectra were used to visually compare the spectral similarity under low and high activation conditions. Not only do the spectral comparisons provide a visual assessment of the similarity, but the spectral comparisons also provide an example of how IS-CID fragment ion spectra can be used to identify substances based on characteristic fragmentation derived from previous MS/MS studies. For example, Fig. 1 shows an exemplar comparison

between the ESI-IS-CID and DART-IS-CID fragment ion spectra for a pure sample of 3-MeOMC under low IS-CID (Fig. 1A and C) and higher IS-CID (Fig. 1B and D) activation conditions. The protonated precursor molecule observed at nominal m/z 194 is the base peak for the low IS-CID fragment ion spectra for both ESI and DART. It should be noted that throughout the presented data there are subtle differences in the accurate mass measurements between ESI-IS-CID and DART-IS-CID fragment ion spectra likely due to the recency of the tune. In addition to the protonated precursor, the low energy ESI-IS-CID spectrum (Fig. 1A) also contains a sodiated adduct at nominal m/z 216 and the presence of fragment ions at nominal m/z 176 and m/z 163. The fragment ions at nominal m/z 176 and m/z 163 are formed through the loss of H_2O and the amine moiety, which are both common fragmentation pathways for *N*-alkylated synthetic cathinones, such as 3-MeOMC [36,37].

At the higher IS-CID activation conditions additional fragmentation is observed, including characteristic fragment ions at nominal m/z 161, m/z 146, and m/z 135. The fragment ion observed at nominal m/z 161 is formed through the loss of a methyl radical from the primary fragment ion at m/z 176 following the proposed fragmentation pathways explained elsewhere [37]. Whereas the loss of a radical from an even-electron species is uncommon, as explained by the even-electron rule [38], this behavior has been observed previously for other synthetic cathinones [37,39,40]. The fragment ions at nominal m/z 146 and m/z 135 are formed through the loss of formaldehyde (CH_2O) and carbon monoxide (CO) from intermediate ions at nominal m/z 176 and m/z 163, respectively. There are also differences in the abundance of smaller fragment ions, such as nominal m/z 132, m/z 118, and m/z 105 between the ESI-IS-CID and DART-IS-CID fragment ion spectra. However, the resulting IS-CID fragment ion spectrum from either technique contains the protonated precursor ion and characteristic fragment ions for 3-MeOMC. Figs. S1–S11 show the remaining spectral comparisons between the ESI-IS-CID and DART-IS-CID fragment ion spectra under low and high activation conditions for the pure designer drugs analyzed in this study.

Whereas Fig. 1 is an example of the similarity between ESI-IS-CID and DART-IS-CID fragment ion spectra for pure compounds, it is also important to consider the impact of mixtures on the resulting IS-CID fragment ion spectra. Fig. 2 shows a comparison of the ESI-IS-CID and DART-IS-CID fragment ion spectra for Mixture 1, which contains α -methyl acetyl fentanyl and fentanyl. This mixture presents a particularly difficult challenge due to the isobaric nature of α -methyl acetyl fentanyl and fentanyl. For example, in the low energy IS-CID fragment ion spectra (Fig. 2A and C) only a single abundant ion is observed at nominal m/z 337, which in this case contains the protonated precursor ions for both α -methyl acetyl fentanyl and fentanyl. The only noticeable difference in the low energy IS-CID fragment ion spectra is the presence of a small sodiated adduct observed at nominal m/z 359 in the ESI-IS-CID spectrum (Fig. 2A).

In comparison, the higher IS-CID activation conditions produce a series of fragment ions that easily enable the identification of at least two fentanyl analogs due to the distribution of fragment ions at nominal m/z 202 and m/z 188. The fragment ion observed at nominal m/z 188 consists of at least three isobaric product ions formed through divergent pathways as described previously [41,42]. Conversely, the fragment ion at m/z 202 is formed through opening of the piperidine ring [43] but shifted 14 Da due to the acetyl group for α -methyl acetyl fentanyl. Likewise, the presence of fragment ions at nominal m/z 119, m/z 105, and m/z 91 provide additional support for the presence of at least two fentanyl analogs as fragment ions at m/z 119 and m/z 91 are associated with α -methyl substitutions, whereas the fragment ion at m/z 105 is not associated with α -methyl substitutions.

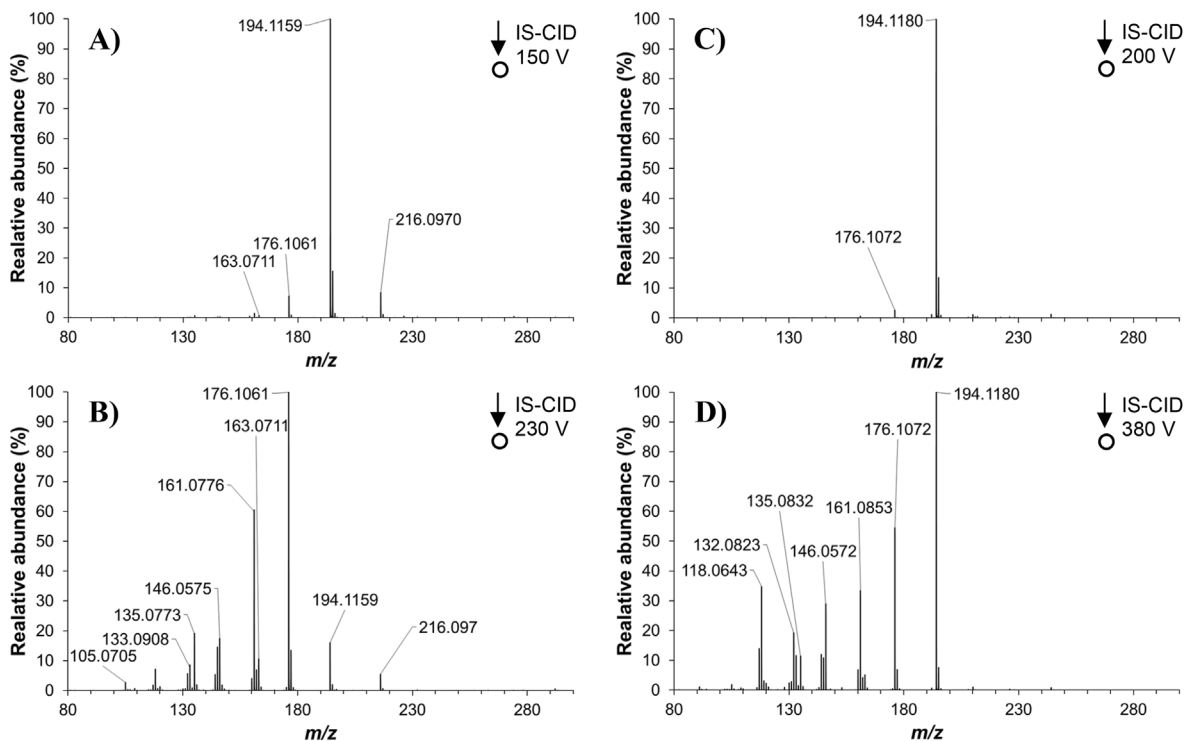


Fig. 1. Comparison of 3-MeOMC fragmentation spectra generated under the following conditions: A) low activation ESI (150 V), B) higher activation ESI (230 V), C) low activation DART (200 V), and higher activation DART (380 V).

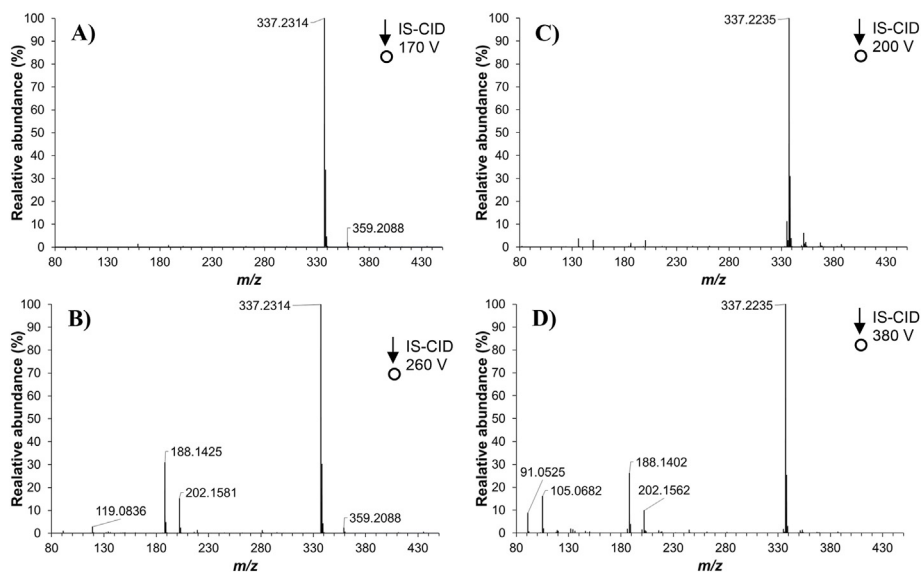


Fig. 2. Comparison of the fragment ion spectra for a mixture of α -methyl acetyl fentanyl and fentanyl (Mixture 1) generated under the following conditions: A) low activation ESI (170 V), B) higher activation ESI (260 V), C) low activation DART (200 V), and higher activation DART (380 V).

The final spectral comparison for ESI-IS-CID and DART-IS-CID fragment ion spectra is Mixture 4, which contains MAB-CHMINACA, α -PVP, and crotonyl fentanyl (Fig. 3). Excluding the presence of the sodiated adduct of MAB-CHMINACA observed at nominal m/z 393, the resulting low energy IS-CID fragment ion spectra contain similar distributions of protonated precursor ions with both ionization techniques (Fig. 3A and C). The higher energy IS-CID fragment ion spectra contain distributions of both the protonated precursor ions and characteristic fragment ions that can be

used for compound identification (Fig. 3B and D). For example, the fragment ions at nominal m/z 189, m/z 161, and m/z 126 correspond with characteristic fragment ions for α -PVP [36,44]. Likewise, the fragment ions at nominal m/z 326, m/z 241, and m/z 145 originate from MAB-CHMINACA [45], and the fragment ion at nominal m/z 188 is a commonly observed fragment ion for fentanyl analogs [41,43], such as crotonyl fentanyl. Even though there are noticeable spectral differences in terms of both relative ion abundance and the presence/absence of specific fragment ions, either higher energy IS-

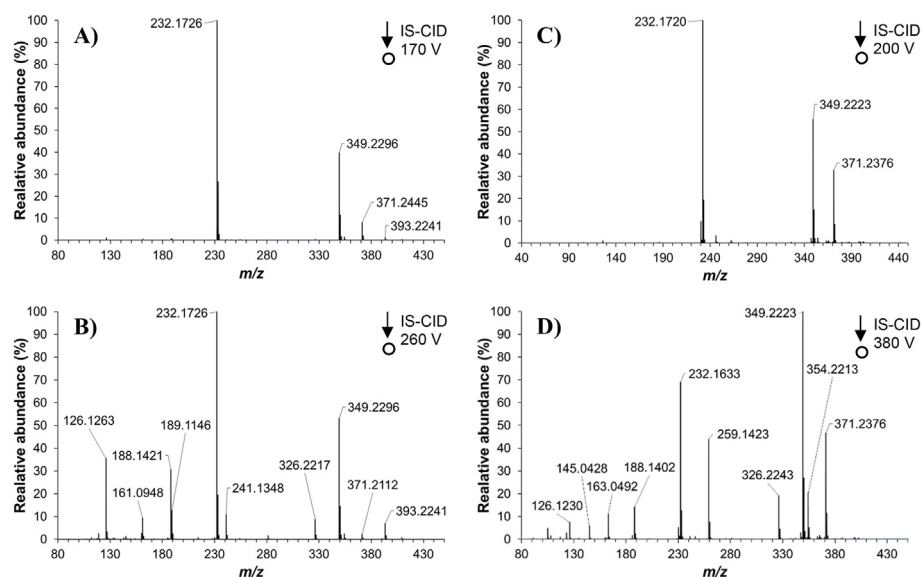


Fig. 3. Comparison of the fragment ion spectra for a mixture of MAB-CHMINACA, α -PVP, and crotonyl fentanyl (Mixture 4) generated under the following conditions: A) low activation ESI (170 V), B) higher activation ESI (260 V), C) low activation DART (200 V), and higher activation DART (380 V).

CID spectrum could be used to identify the presence of MAB-CHMINACA, α -PVP, and crotonyl fentanyl through the combination of protonated precursor ions and characteristic fragment ions. Figs. S12–S14 provide the spectral comparisons between the ESI-IS-CID and DART-IS-CID fragment ion spectra under low and high activation conditions for the remaining simulant mixture samples.

3.2. Pearson product-moment correlations (PPMCs)

PPMCs were used to statistically assess the similarity of the ESI-IS-CID and DART-IS-CID fragment ion spectra. When comparing the highest PPMC coefficient within each comparison for a given set of activation conditions across all 12 compounds in this study, the average highest PPMC coefficient with a 95% confidence interval was 0.9295 ± 0.0031 . These results indicate that under the most favorable conditions, there is a high degree of spectral similarity between ESI-IS-CID and DART-IS-CID fragment ion spectra. Readers who may be interested in the use of PPMC to identify the optimum activation conditions for spectral comparison should see our previous work [25]. When broken down by compound class, the resulting average highest PPMC coefficients with a 95% confidence interval were 0.9295 ± 0.0031 , 0.9440 ± 0.0229 , and 0.9641 ± 0.0391 for the synthetic cannabinoids, synthetic cathinones, and fentanyl analogs, respectively. In all cases, the per class averages demonstrate that a high degree of spectral similarity between ESI-IS-CID and DART-IS-CID fragment ion spectra can be achieved. It should be noted that in addition to any class-specific fragmentation differences, the data vector length also effects the resulting PPMC coefficients. The data vector length was not held constant between classes to avoid comparisons between adducts formed with ESI that are absent from DART ionization. However, differences in data vector length do not appear to dominate this comparison as the synthetic cathinones had the shortest vector length on average, but the synthetic cannabinoids had the lowest 95% confidence interval for the highest PPMC coefficients. Table S1 provides a summary of the PPMC coefficient results broken down by overall average, compound class, and individual compound. The highest average PPMC coefficient for an individual compound was 0.9713 ± 0.0254 for alfentanil, whereas the lowest average PPMC coefficient for the highest pairwise comparison was

0.8688 ± 0.0431 for AB-PINACA. Interestingly, most of the highest spectral comparisons for the low, medium, and high DART-IS-CID fragmentation occurred between fragmentor voltages of 150 V and 240 V for the ESI-IS-CID data.

3.3. NIST DART-MS Forensics Database and DIT

3.3.1. DART-IS-CID fragment ion spectra

The NIST DART-MS Forensics Database and DIT is a freely available resource provided by NIST to assist forensic laboratories with mixture interpretation of IS-CID fragment ion spectra. However, the database and DIT were developed using only spectra collected on JEOL AccuTOF high-resolution mass spectrometers coupled with DART ionization. Whereas DART is most often combined with JEOL AccuTOF mass spectrometers, this will not be the only configuration being used for seized drug casework, due to the ability to mount the DART source onto nearly any type of mass spectrometer with the addition of a Vapor interface. Thus, it is important to understand the applicability of the NIST DART-MS Forensics Database and DIT for data collected with DART ionization on other mass spectrometers, such as the Agilent 6530 Q-TOF used in this study.

Table 1 provides a summary of the DIT results for the analysis of a mixture containing AM-2201, 4-CEC, and alfentanil (Mixture 3) collected under low IS-CID activation conditions (i.e., a 200 V fragmentor voltage) with DART ionization. Only selected metrics from the DIT are provided in Table 1. The DIT provides a series of spectral similarity metrics, which have been described extensively elsewhere [33,34]. Briefly, the reverse match factor (RevMF) is the spectral similarity between the query and reference spectrum ignoring any ions present in the unknown that are absent in the reference spectrum. The $\Delta m/z$ metric provides the difference between the measured m/z value and the theoretical exact mass. Similarly, the isotope ratio difference (IRD) metric provides the similarity between the observed ratio of the protonated precursor ion and its major isotope to the theoretical ratio, with values closer to zero indicating higher similarity. The match type indicates what type of ion from the reference spectrum matches with the identified target. Match types can include protonated molecules (PM), PM isotope, base peak, base peak isotope, and major fragment ion. An

Table 1

DIT search results for Mixture 3 using the DART-IS-CID fragmentation data collected under low IS-CID activation conditions.

Target #	Target m/z	Relative Intensity (%)	Compound	RevMF	$\Delta m/z$	IRD	Match Type	Ground Truth
1	417.2610	100.0	Alfentanil	0.832	−0.0004	0.074	PM	Present
2	360.1758	74.9	AM-2201	0.935	−0.0006	0.033	PM	Present
3	418.2637	30.1	Alfentanil	0.832	−0.0011	0.074	PM (Isotope)	Present
4	361.1790	21.9	AM-2201	0.935	−0.0007	0.033	PM (Isotope)	Present
5	212.0833	20.1	4-CEC	0.849	−0.0009	−0.031	PM	Present
7	266.1610	6.3	Mirtazapine	0.770	−0.0047	−0.184	PM	Absent
8	214.0783	5.8	4-CEC	0.849	−0.0030	−0.031	PM (Isotope)	Present

*PM = Protonated molecule.

*PM (Isotope) = Isotope of the protonated molecule.

additional column was added for the ground truth information about the presence or absence of the identified substance in the mixture. Finally, the compound(s) listed in each DIT summary table correspond to either the ground truth compound if it was identified for a given target or the highest-ranking match based on the RevMF score.

As anticipated, under low IS-CID activation conditions the formation of protonated precursor ions dominate the DART-IS-CID fragment ion spectrum (Fig. S13). The DIT identified eight targets in the low energy spectrum using a target threshold of 5% abundance and an m/z tolerance of 5 mDa. These thresholds were chosen to reduce the influence of background ions, which are commonly present with ambient ionization sources such as DART, and to reduce the potential for false positive identifications by using a relatively narrow m/z tolerance. It should be noted that DIT results will vary based on the chosen target threshold and m/z tolerance. Six of the eight identified targets belong to compounds known to be present in the mixture with the match types of PM and PM isotope. Two targets did not correspond to the known ground truth mixture composition. Target 6 was an ion at m/z 359.1679 that was present at 7.5% abundance, but there were no matches in the database, which is why Target 6 is not reported in Table 1. This approach of excluding targets that do not contain a match in the database was used for the remainder of the manuscript. Target 7 is the lone potential match that was not present in the mixture. However, Target 7, identified as mirtazapine, had the lowest RevMF and the most disparate $\Delta m/z$ and IRD metrics, which has been previously proposed as an approach to identify library matches not likely to be present in the sample [46]. Importantly, even if mirtazapine was not recognized as a potential false positive, mirtazapine is not a controlled substance, and the three controlled substances that were present in the mixture were all correctly identified as potential matches.

The summary of the DIT results for the DART-IS-CID fragmentation data collected with higher IS-CID activation conditions (i.e., a

380 V fragmentor voltage) is provided in Table 2. In total, 25 targets were identified for Mixture 3, but only 11 targets contained a match within the database. This time, 8 of the 11 targets correspond to the three known compounds in Mixture 3, including the match types of PM, PM isotope, and major fragment ion. The three remaining potential matches all corresponded to base peak or base peak isotope match types for tryptamine and gramine based on the presence of relatively small fragment ions (i.e., less than m/z 150). The resulting RevMF scores being lower than the remaining targets and the base peak match type for Target 4 and Target 11 is a potential mechanism to identify these matches as false positives given the measured abundances of 30.1% and 18.4%, respectively. Again, neither of these compounds are controlled substances and the three controlled substances present were correctly identified as being present in the mixture. Tables S2–S9 contain the remaining summaries of the DIT results for the other four simulant mixtures analyzed with DART-IS-CID in this study. In all cases, at least one of the known samples was identified using a minimum of the PM and PM isotope with low IS-CID conditions and an additional major fragment ion or base peak with higher IS-CID conditions.

3.3.2. ESI-IS-CID fragment ion spectra

With the seized drug community beginning to adopt single-stage mass spectrometers coupled with soft ionization sources, such as LC-ESI-MS instrumentation, there is a need to understand the comparability of ESI-IS-CID fragment ion spectra to the DART-IS-CID spectra contained within the NIST DART-MS Forensics Database. In sections 3.1 and 3.2, the authors demonstrated that it is possible to generate similar DART-IS-CID and ESI-IS-CID fragment ion spectra. However, it is also imperative to understand the effectiveness of using the NIST DART-MS Forensics Database and DIT with ESI-IS-CID fragment ion spectra for the screening of seized drug evidence. The same 5% abundance target threshold and 5 mDa m/z tolerance values used for the DART-IS-CID DIT analysis were also applied to the ESI-IS-CID DIT analysis.

Table 2

DIT search results for Mixture 3 using the DART-IS-CID fragmentation data collected under higher IS-CID activation conditions.

Target #	Target m/z	Relative Intensity (%)	Compound	RevMF	$\Delta m/z$	IRD	Match Type	Ground Truth
1	360.1753	100.0	AM-2201	0.936	−0.0011	0.007	PM	Present
2	212.0831	74.3	4-CEC	0.916	−0.0011	0.000	PM	Present
3	417.2602	68.9	Alfentanil	0.935	−0.0012	0.036	PM	Present
4	144.0801	30.1	Tryptamine	0.868	−0.0012	N/A	Base Peak	Absent
6	361.1787	26.7	AM-2201	0.936	−0.0010	0.007	PM (Isotope)	Present
7	155.0488	25.4	AM-2201	0.936	−0.0009	0.007	Major Fragment Ion	Present
8	214.0803	23.8	4-CEC	0.916	−0.0010	0.000	PM (Isotope)	Present
11	130.0648	18.4	Gramine	0.883	−0.0009	N/A	Base Peak	Absent
12	418.2632	18.2	Alfentanil	0.935	−0.0016	0.036	PM (Isotope)	Present
15	268.1763	15.1	Alfentanil	0.935	−0.0010	0.036	Major Fragment Ion	Present
17	131.0706	11.2	Gramine	0.883	−0.0016	N/A	Base Peak (Isotope)	Absent

*PM = Protonated molecule.

*PM (Isotope) = Isotope of the protonated molecule.

The DIT results for Mixture 5 using ESI-IS-CID with low activation conditions (i.e., a 170 V fragmentor voltage) are provided in Table 3. Eight potential targets were identified as matches, with seven of the eight targets matching either the PM or PM isotope for the four compounds known to be present in the mixture. Under these low IS-CID conditions, the protonated precursor ions dominate the spectrum (Fig. S14), although not all protonated precursor ions were overly abundant. For example, the protonated precursor ion of AB-PINACA was only present at 6.8% abundance, demonstrating the effect of competitive ionization for mixture analysis without prior chromatographic separation. Nonetheless, all four compounds present were identified, and the only target identified by the DIT that did not belong to these compounds was 3-chlorophenmetrazine (3-CPM). A base peak match type for 3-CPM was identified by the DIT; however, without the corresponding PM this potential false positive can be quickly eliminated based on the presence of the chlorine within the molecular structure and known isotopic distribution for chlorine. Likewise, as described previously by Sisco et al. the $\Delta m/z$ metric can be used to quickly identify a potential library match that is not likely to be in the mixture based on mass drift in the opposite direction of all other identified matches [46]. In this case, the 3-CPM has a $\Delta m/z$ value of -0.0030 and all other potential matches have $\Delta m/z$ values ranging from 0.0005 to 0.0010 . Furthermore, even if an analyst was not able to eliminate 3-CPM based on the information provided by the DIT, 3-CPM is not scheduled and thus, would not be of concern for further confirmatory analysis.

The most interesting outcome from the DIT analysis of the ESI-IS-CID fragment ion spectra occurred for Mixture 5 with the higher IS-CID activation conditions (i.e., a 380 V fragmentor voltage) shown in Table 4. The DIT returned 14 potential targets with matches within the database. Of these 14 targets, 7 correspond with the PM, PM isotope, or base peak of one of the known compounds. However, several of the incorrect matches were correct in terms of the appropriate class identification which is of the utmost importance for screening techniques. For example, Targets 2 and 13 were potential hits for the base peak and base peak isotope for cyclohexyl fentanyl. In fact, all ten potential hits were fentanyl analogs for both targets, which makes sense given that the fragment ion at m/z 188.1439 is the base peak for many fentanyl analogs [41–43]. From that perspective, Targets 2 and 13 were also correct in terms of the compound class identification since both fentanyl and crotonyl fentanyl were present in Mixture 5. Likewise, Target 12 indicated a major fragment ion for bk-DMBDB, which is a constitutional isomer of pentylone and produces the same fragment ion at m/z 149.0239. Finally, Target 5 matched with 5-methoxy-AMT with a base peak ion at m/z 188.1076. Given the ground truth knowledge about the composition of the mixtures, it is far more likely that the target at m/z 188.1076 is the fragment ion with the elemental composition $C_{12}H_{14}NO^+$ with a theoretical exact

mass of m/z 188.1075 formed from the sequential loss of H_2O and formaldehyde (CH_2O) from the protonated precursor ion for pentylone [37]. The remaining three identified targets that could be considered false positives are all non-controlled substances and thus, are not of consequence for seized drug analysts. The supplemental material contains the remaining summaries for the ESI-IS-CID DIT results for the other four simulant mixtures (Tables S10–S17). The PM for each known compound was identified as a match in the low IS-CID spectrum for all five mixtures, although the PM isotope was missing for at least one compound in the low IS-CID spectrum for three of the five mixtures. The higher IS-CID data provided additional information in terms of major fragment ions and/or base peak ions for all five mixtures, although the presence of major fragment ions led to the loss of the PM and PM isotope in three of the five mixtures.

3.3.3. Effectiveness of NIST DART-MS Forensics Database and DIT

Overall, the effectiveness of the NIST DART-MS Forensics Database and DIT were relatively conserved between the DART-IS-CID and ESI-IS-CID data. In all cases, at least one of the known compounds present in the mixture was identified using a minimum of the PM and PM isotope with the low IS-CID conditions and an additional base peak or major fragment ion with the higher IS-CID conditions. However, the number of identified targets and match types are related to the IS-CID activation conditions. For example, the low IS-CID spectrum is most useful for identifying the PM and PM isotope match types depending on the protonated precursor ion abundance, whereas the higher IS-CID spectrum is more useful for identifying base peak or major fragment ion match types. In practice, this means that the higher energy IS-CID spectrum may provide more false positives through potential matches to base peak and major fragment ions, but the higher IS-CID spectrum may also provide additional support for the presence of the correct compounds through the identification of base peak or major fragment ions. Understanding how to exclude potential false positive identifications through metrics such as $\Delta m/z$ and IRD and recognizing common major fragment ions that may be misidentified as base peak ions are important aspects to reducing false positive identifications. For example, there were several instances where the DIT identified a potential match to the base peak of a substance that was not present in the mixture but was within the m/z tolerance window of a major fragment ion for a compound that was present in the mixture. Examples include m/z 188.1076 for 5-methoxy-AMT instead of m/z 188.1075 ($C_{12}H_{14}NO^+$) for pentylone (DART-IS-CID Mixture 5), m/z 109.0541 for 4-fluoroamphetamine instead of m/z 109.0448 ($C_7H_6F^+$) for AB-FUBINACA (DART-IS-CID Mixture 2), and m/z 241.1355 for APP-CHMINACA instead of m/z 241.1335 ($C_{15}H_{17}N_2O^+$) for MAB-CHMINACA (ESI-IS-CID Mixture 4) [45]. In these three cases, the fragment ions were present at abundances of 34%, 5%, and 10%, respectively even though they were matched to

Table 3

DIT search results for Mixture 5 using the ESI-IS-CID fragmentation data collected under low IS-CID activation conditions.

Target #	Target m/z	Relative Intensity (%)	Compound	RevMF	$\Delta m/z$	IRD	Match Type	Ground Truth
1	236.1292	100.0	Pentylone	0.504	0.0005	0.028	PM	Present
2	349.2288	75.7	Crotonyl fentanyl	0.764	0.0008	0.036	PM	Present
3	337.2287	60.0	Fentanyl	0.843	0.0007	0.030	PM	Present
4	350.2321	21.6	Crotonyl fentanyl	0.764	0.0008	0.036	PM (Isotope)	Present
5	237.1325	16.8	Pentylone	0.504	0.0005	0.028	PM (Isotope)	Present
6	338.2321	16.1	Fentanyl	0.843	0.0008	0.030	PM (Isotope)	Present
7	331.2144	6.8	AB-PINACA	0.026	0.0010	0.012	PM	Present
8	218.1186	5.0	^a 3-CPM	0.900	-0.0030	-0.031	PM (Isotope)	Absent

*PM = Protonated molecule.

*PM (Isotope) = Isotope of the protonated molecule.

^a3-CPM = 3-Chlorophenmetrazine.

Table 4

DIT search results for Mixture 5 using the ESI-IS-CID fragmentation data collected under higher IS-CID activation conditions.

Target #	Target m/z	Relative Intensity (%)	Compound	RevMF	$\Delta m/z$	IRD	Match Type	Ground Truth
1	236.1292	100.0	Pentylone	0.725	0.0005	0.017	PM	Present
2	188.1431	68.4	^a Cyclohexyl fentanyl	0.884	-0.0008	N/A	Base peak	Absent
3	349.2289	66.9	Crotonyl fentanyl	0.859	0.0009	0.024	PM	Present
4	337.2288	52.8	Fentanyl	0.707	0.0008	0.020	PM	Present
5	188.1076	34.0	^b 5-methoxy AMT	0.817	0.0001	N/A	Base peak	Absent
8	350.2324	18.2	Crotonyl fentanyl	0.859	0.0011	0.024	PM (Isotope)	Present
9	237.1328	15.8	Pentylone	0.725	0.0008	0.017	PM (Isotope)	Present
10	135.0451	14.4	MDBP	0.840	0.0005	N/A	Base peak	Absent
11	338.2325	13.6	Fentanyl	0.707	0.0012	0.020	PM (Isotope)	Present
12	149.0247	10.7	^c Bk-DMBDB	0.789	0.0008	0.017	Major Fragment ion	Absent
13	189.1467	10.4	^d Cyclohexyl fentanyl	0.884	-0.0006	N/A	Base peak (Isotope)	Absent
14	215.1191	10.3	AB-PINACA	0.837	0.0007	N/A	Base peak	Present
19	189.1040	5.4	Antipyrine	0.783	0.0012	-0.119	PM	Absent
20	159.0688	5.0	Cimetidine	0.306	-0.0016	N/A	Base peak	Absent

*PM = Protonated molecule.

*PM (Isotope) = Isotope of the protonated molecule.

^aThe fragment ion at m/z 188.1439 is the base peak for many fentanyl analogs including fentanyl and crotonyl fentanyl.^bMore likely to be m/z 188.1075 for the $C_{12}H_{14}NO^+$ fragment ion from pentylone.^cBk-DMBDB is a constitutional isomer of pentylone and both produce the fragment ion at m/z 149.0239.^dThe fragment ion at m/z 189.1461 is the base peak isotope for many fentanyl analogs including fentanyl and crotonyl fentanyl.

the base peak for the incorrect compound.

Given that the NIST DART-MS Forensics Database and DIT were developed to be combined with DART-MS screening for seized drugs, it is more important to minimize false negatives even at the potential expense of false positives because additional confirmatory testing will help eliminate false positive identifications. However, based on the results of this study, it may be possible to develop criteria to help reduce false positives without increasing false negatives. The following are the proposed criteria to accomplish this goal: 1) collection and analysis of IS-CID fragment ion spectra at a minimum of low and high activation conditions; 2) at least one database match for a given compound in both the low and high IS-CID spectra; and 3) there must be a match to at least the PM in the low energy IS-CID spectrum. When these criteria are applied to the five simulant mixtures analyzed in this study, across the two collection methods, the ground truth compound was identified in all cases except one (i.e., one false negative) and there was only a single incorrect identification (i.e., one false positive). The false negative was for Mixture 2 using DART-IS-CID (Table S5) due to the absence of any match to AB-FUBINACA in the higher energy IS-CID spectrum, although there were three identified matches to AB-FUBINACA or an isomer of AB-FUBINACA in the low energy IS-CID spectrum (Table S4) and a misidentified major fragment ion at m/z 109.0451 in the higher energy IS-CID spectrum (Table S5). The one false positive was for Mixture 2 using ESI-IS-CID due to the presence of a match for the PM of 5-APB in the low energy IS-CID spectrum (Table S12) and the presence of a match for the PM and PM isotope of 5-APB in the higher energy IS-CID spectrum (Table S13). However, the PM match for 5-APB at m/z 176.1082 is within the m/z tolerance window of the correct major fragment ion at m/z 176.1075 ($C_{11}H_{14}NO^+$) for 3-MeOMC, which was present in the mixture. Finally, when the proposed criteria were tested against the analysis of a negative control (i.e., the average blank spectrum), there were no false positive identifications, which demonstrates the effectiveness of the proposed criteria for minimizing false negatives without increasing false positives.

Even though the results of this study provide compelling support for the potential compatibility of the NIST DART-MS Forensics Database and DIT to other soft ionization sources on single-stage mass spectrometers other than the JEOL AccuTOF mass spectrometer, it is important to highlight several limitations of this study. First, the presented results were collected with a target threshold of

5% and an m/z tolerance of 5 mDa. Altering these thresholds will provide different results in terms of potential false positives and false negatives. As the target threshold is lowered, additional targets are considered, which can increase the length of time and difficulty for completing the data analysis. Likewise, expanding the m/z tolerance will allow for additional target ions to be considered. A second limitation is that the five simulant mixtures did not contain any diluents or adulterants and only included 12 representative designer drugs from the synthetic cannabinoids, synthetic cathinones, and fentanyl analog compound classes. Finally, the proposed criteria were not assessed against additional data sets and as such, may only be applicable to the compounds and/or conditions used for the DIT analysis in this study. Our ongoing research will help address these limitations through the analysis of realistic casework mixtures, incorporating additional seized drugs, and investigating the impact of the target threshold and m/z tolerance on the resulting efficiency of the NIST DART-MS Forensics Database and DIT.

4. Conclusions

This manuscript demonstrates the similarity between ESI-IS-CID and DART-IS-CID fragment ion spectra through the combination of spectral comparisons and PPMCs. Spectral comparisons demonstrated similar distributions of fragment ions between ESI-IS-CID and DART-IS-CID activation under varying activation conditions, as well as the presence of previously characterized fragment ions for the representative designer drugs analyzed in this study. Likewise, the 95% confidence interval for the average highest PPMC coefficient across the 12 designer drugs in this study was 0.9295 ± 0.0031 . Finally, the DART-IS-CID and ESI-IS-CID fragment ion spectra collected on an Agilent 6530 Q-TOF mass spectrometer produced similar identification results for the five simulant mixtures when uploaded to the DIT and searched against the NIST DART-MS Forensics Database. With the increasing reliance on IS-CID with single-stage mass spectrometers in routine seized drug casework, providing evidence for the potential compatibility of the NIST DART-MS Forensics Database and DIT with non-DART AccuTOF instrumentation is a significant contribution from this work. Even with a limited number of samples, our results demonstrate the potential applicability of the NIST DART-MS Forensics Database and DIT to help with the interpretation of seized drug mixtures based

on IS-CID fragment ion spectra collected with non-DART AccuTOF instrumentation.

CRediT authorship contribution statement

Alleigh N. Couch: Investigation, Formal analysis, Writing – original draft. **Jared Sharp:** Formal analysis, Writing – review & editing. **J. Tyler Davidson:** Conceptualization, Methodology, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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